

Metabolic buckling of the growing spine

Sayuj Krishnan S^{*1}

¹Yashoda Hospitals, Hyderabad, India

Abstract

Living systems routinely maintain structure against gravity, yet the physical principle underlying this ability remains unexplained. We show that Adolescent Idiopathic Scoliosis (AIS) is a “Metabolic Buckling” event caused by a mismatch between the scaling laws of mechanical demand and metabolic supply. Using a Cosserat rod model coupled to a developmental information field, we identify a critical “Energy Deficit Window” at spinal lengths $L > 0.35$ m where the thermodynamic cost of straightness exceeds metabolic capacity by over 40%. Structural analysis of 23 key proteins reveals that mechanosensors required for alignment (e.g., Vimentin, anisotropy 7.5) are thermodynamically expensive, while the metabolic master regulator (PPARGC1A) is intrinsically disordered. The decoupling of anisotropic stiffness from isotropic growth under energy deficit inevitably leads to rotational instability. These results reclassify scoliosis not as a bone disease, but as a protective metabolic strategy to reduce the thermodynamic cost of standing.

Life on Earth has evolved within the unyielding context of a constant gravitational field, which acts not merely as a mechanical load but as a ubiquitous morphogenetic cue [1]. For sessile structures, gravity dictates a simple equilibrium: a beam clamped at one end will sag into a passive C-shape. In vertebrates, however, the spine maintains a complex, metastable S-shaped profile (lordosis and kyphosis) against gravity [2]. We propose that this “Biological Countercurvature” is not a passive equilibrium but a Thermodynamic Standing Wave—a dissipative structure maintained by the continuous expenditure of metabolic energy against the gravitational field.

The Scaling Catch-22 defines the central conflict of vertebrate growth. From a physics perspective, the spine acts as an Euler column where the critical buckling load scales as $1/L^2$. To prevent buckling, the flexural stiffness must increase dramatically. We show that the metabolic power (P_{counter}) required to maintain straightness scales as L^4 , whereas the biological machinery supplying this energy scales only as L^2 [3, 4]. This divergence creates a critical Energy Deficit Window during the rapid adolescent growth spurt. We calculate the crossover point where metabolic demand exceeds supply and find it occurs at a spinal length $L_{\text{crit}} \approx 0.35$ m. At $L = 0.45$ m, typical of mid-adolescence, the metabolic deficit reaches 45.7%, leaving the spine energetically insolvent.

Vector-Scalar Mismatch emerges as the failure mode during this deficit. Healthy morphogenesis requires the integration of directional Vector signals (gravity sensing via proprioception) with isotropic Scalar growth signals (IGF-1/GH). We simulate this coupling using a 3D Cosserat rod model driven by an Information-Elasticity Coupling (IEC) field. In the Cooperative Regime ($\chi_\kappa \approx 0.05$), information and gravity balance to produce a stable S-curve. However, when metabolic supply fails, the expensive Vector system collapses while the Scalar growth drive persists. Our simulations show that this decoupling triggers a supercritical bifurcation: small asymmetries ($\varepsilon_{\text{asym}} < 1\%$) that are normally suppressed are amplified into macroscopic helical deformities, reproducing the 3D geometry of scoliosis with Cobb angles $> 15^\circ$.

^{*}hellodr@drsayuj.info

The Protein Cost Landscape provides the molecular basis for this failure. We hypothesized that the weak links are the proteins responsible for sensing curvature. Using AlphaFold-derived metrics for 23 key proteins, we compute a Thermodynamic Cost (Anisotropy $\times$ Residues). The analysis reveals a stark Demand-Supply Anisotropy Gap of 34%. Demand-side mechanosensors are structurally highly anisotropic (mean 3.32) and expensive to maintain. Vimentin, the intermediate filament acting as a strain gauge [5], has an extreme anisotropy of 7.5, making it the most vulnerable to energy depletion. Conversely, the supply-side regulator PPARGC1A (PGC-1 α) is highly disordered (62% disorder) and prone to degradation. This creates a VIM Cascade: energy stress leads to the collapse of high-anisotropy sensors (Vimentin $\rightarrow$ Lamin A/C $\rightarrow$ Piezo2), blinding the spine to its own curvature just when growth is fastest.

Sexual Dimorphism and the Metabolic Depth hypothesis explain the 10:1 female prevalence of severe scoliosis. Our model indicates that females face a narrower but deeper Energy Deficit Window. With earlier peak height velocity and a higher fat-to-muscle mass ratio, the peak deficit ratio in girls is estimated at $R_{\text{peak}} \approx 2.7$ versus 2.4 in boys. Furthermore, lower basal expression of PPARGC1A in female paraspinal muscles [6] shifts L_{crit} to shorter lengths, initiating vulnerability earlier in the growth curve.

Six Mechanisms of Supply Failure drive the system into the deficit window. Beyond simple caloric insufficiency, we identify: (1) A mitochondrial capacity ceiling set by fragile PPARGC1A; (2) Vascular supply lag leading to local hypoxia and a glycolytic shift; (3) Circadian desynchrony (Spinal Jetlag) where clock proteins like ARNTL (Anisotropy 3.3) fail to synchronize matrix repair with loading cycles [?]; (4) Micronutrient deficiency (NAD $^{+}$ precursors) blinding the SIRT1 energy gauge; (5) The Modern Mismatch of increased height; and (6) A recursive supply constraint where building the supply machinery itself consumes the limited energy. While direct flux measurements are invasive, the scaling laws provide a robust theoretical bound confirming that these supply failures inevitably precipitate structural collapse.

Metabolic Buckling thus redefines AIS. It is not a random genetic error but a predictable system failure. The deformity is, in fact, a mechanism of thermodynamic survival: by buckling into a configuration that lowers the center of mass or reduces the active moment required to stay upright, the organism sheds the Anisotropy Tax it can no longer afford. This framework suggests that therapeutic interventions should shift from purely mechanical bracing to metabolic rescue—targeting mitochondrial biogenesis, circadian synchronization, and micronutrient sufficiency to close the Energy Deficit Window.

Methods

Cosserat Rod Simulation

We utilized PyElastica [7], an open-source Python implementation of Cosserat rod theory, to model the spine as a continuous elastic rod. The rod was discretized into $N = 50 - 100$ elements. The “Biological Countercurvature” was implemented via an Information-Elasticity Coupling (IEC) term that modifies the local rest curvature $\kappa^0(s)$ based on a scalar information field $I(s)$ representing HOX patterning. The field was modeled as a bimodal Gaussian distribution corresponding to cervical and lumbar lordosis.

Energy Deficit Calculation

The metabolic cost of countercurvature (P_{counter}) was defined as proportional to $L^2 \int (\kappa_{\text{IEC}} - \kappa_{\text{passive}})^2 ds$. Supply capacity (S_{proprio}) was modeled scaling as L^2 (surface-limited) or L^3 (volume-limited). The deficit ratio $R = P_{\text{counter}}/S_{\text{proprio}}$ was computed across spinal lengths $L \in [0.2, 0.6]$ m.

AlphaFold Structural Analysis

We retrieved predicted structures for 23 proteins from the AlphaFold Protein Structure Database [?]. Structural anisotropy was calculated via the Gyration Tensor of the atomic coordinates. Disorder fraction was computed based on pLDDT scores < 70 . The “Thermodynamic Cost” was defined as Anisotropy $\times$ Number of Residues.

References

- [1] D’Arcy Wentworth Thompson. *On Growth and Form*. Cambridge University Press, Cambridge, 1917.
- [2] B. Latimer. The evolutionary biomechanics of the human spine. *Journal of Anatomy*, 207: 617–617, 2005.
- [3] Geoffrey B West, James H Brown, and Brian J Enquist. A general model for the origin of allometric scaling laws in biology. *Science*, 276(5309):122–126, 1997. doi: 10.1126/science.276.5309.122.
- [4] D. F. S. Rolfe and G. C. Brown. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiological Reviews*, 77(3):731–758, 1997.
- [5] S. L. Wuest et al. Vimentin intermediate filaments act as a gravitational strain gauge in eukaryotic cells. *Nature Microgravity*, 9, 2025. doi: 10.1038/s41526-025-00123-x.
- [6] C. Handschin et al. Paraspinal muscle atrophy in microgravity: molecular mechanisms. *Cell Metabolism*, 36, 2024. Mechanisms of anti-gravity muscle loss.
- [7] Arman Tekinalp, Mattia Gazzola, et al. Pyelastica: Open-source software for the simulation of assemblies of slender, one-dimensional structures using cosserat rod theory, 2023. URL <https://doi.org/10.5281/zenodo.7658872>.